Error robustness of doublet and triplet genetic codes

Hannah E. Ribbeck[1], J. Paudyal[1], Boggavarapu Kiran[1,*]

1. Department of Chemistry and Physics, McNeese State University, Lake Charles, LA 70609, USA

*Boggavarapu Kiran
Department of Chemistry and Physics
McNeese State University
Lake Charles, LA 70609, USA
Email: kiran@mcneese.edu

Keywords

Genetic code; Codon length; Translation error; Hamming graph; Edge-isoperimetric inequality; Wobble position

Abstract

Whether a triplet genetic code protects proteins against translation errors better than a doublet code has not been tested by comparing the two architectures directly. We define the expected physicochemical cost of a single-nucleotide misreading per translated residue, at fixed amino-acid usage, on the codon Hamming graph, and compute its minimum over all codes of each length. Codon length alone buys nothing: an edge-isoperimetric bound that does not depend on length sets the minimum without physicochemical distances, and with distances the optimal doublet and triplet codes for ten amino-acid classes coincide under uniform usage and differ by 1.6% under proteome usage. A longer code gains only by making its noisiest position synonymous, so its advantage is the cost of the channel it discards. At in vitro misreading rates that channel is position 3, which the standard code silences; the best quadruplet code we found gains 2 to 6% over the best triplet code, for a one-third longer message, which is why codons stop at three. The robustness benefit of the triplet architecture is therefore set by position-specific misreading rates, not by codon length.

Introduction

Ribosomes misread about one codon in 10^3 to 10^4 (Kramer and Farabaugh 2007; Drummond and Wilke 2008; Landerer et al. 2024). The cost of a misreading depends on which amino acid replaces which, and the standard genetic code (SGC) arranges its assignments so that single-nucleotide neighbours tend to encode similar amino acids (Woese 1965; Sonneborn 1965). Haig and Hurst (1991) turned this observation into a test by scoring the SGC against codes in which the amino acids are permuted among the synonymous codon blocks, and Freeland and Hurst (1998) found the SGC better than all but about one in a million such codes. Later work showed that the percentile depends on the amino-acid property scored (Koonin and Novozhilov 2009) and on the null model (Rozhoňová and Payne 2021), and that local optimization of random codes reaches comparable robustness (Novozhilov et al. 2007). Putative primordial codes whose third position is fully redundant are nearly optimal among permutations of their assignments (Novozhilov and Koonin 2009), and the robustness of a code has been posed as an optimal partition of the codon graph (Błażej et al. 2022).

All of these tests fix codon length. Codons are triplets because 16 doublets cannot name 20 amino acids and a stop signal, and triplets are the shortest words that can (Crick 1968). That arithmetic says nothing about error robustness. A triplet code has more positions at which a misreading can occur and more codons per amino acid with which to absorb it, and the two effects pull in opposite directions. Crick's wobble hypothesis (1966) supplies the qualitative picture, in which the third position is read loosely and the code places its synonyms there, but the net effect of codon length on the cost of misreading has not been computed.

The question has a clean form. For a fixed set of amino acids used at fixed frequencies, what is the lowest expected cost of a single-nucleotide misreading per translated residue that any doublet code can achieve, what is the lowest that any triplet code can achieve, and where does the SGC sit? Comparing z-scores of observed codes against architecture-specific nulls does not answer it, because each z-score measures an assignment against its own ensemble and the ensembles differ in size and structure. We answer it by computing the minima themselves, exactly where a combinatorial bound applies and by certified search where it does not. We then add position-specific misreading rates, because the only way a longer code can help turns out to depend on them, and use the same framework to ask why codons are not four nucleotides long.

Results

The per-residue cost of misreading

We represent a code of length n as a labelling of the 4^n codons of the Hamming graph $H(n, 4)$, in which two codons are adjacent if they differ at one position (fig. 1a). Each of k amino-acid classes a occupies a block of n_a codons and is used at frequency f_a. The expected cost of misreading per translated residue is

$$D_w = \Sigma_a f_a \, (1/n_a) \, \Sigma_{c \in a} \, \Sigma_p \, w_p \, \Sigma_{c' \in N} p_{(c)} \, d(a, a(c')),$$

where $N_p(c)$ is the set of sense codons that differ from c only at position p, w_p is the relative misreading rate at position p, and d is the physicochemical distance between classes, zero for synonymous neighbours. With $w_p = 1$ and $f_a = n_a/N$, D_w reduces exactly to the codon-weighted error load of Tlusty (2007), which we call D. Fixing f is what makes codes of different length comparable: every code of every length translates the same proteome. Codon multiplicity in the SGC tracks amino-acid usage (Seo et al. 2026), which is why we report proteome usage alongside uniform usage.

Distances come from eight principal components of 20 standardized amino-acid descriptors taken from AAindex or computed from side-chain structure (97.5% of variance), each component rescaled to unit variance (table S2; the effect of this rescaling is reported below). For the architecture comparison we use the ten reduced-alphabet classes of Murphy et al. (2000), each represented by one member (Materials and Methods), because a doublet code cannot encode 20.

Codon length does not change the achievable cost

A doublet code embeds in a triplet code without changing its cost. Given a doublet labelling L_2, the triplet labelling $L_3(xyz) = L_2(xy)$ makes position 3 fully synonymous. Every codon keeps its doublet neighbours at positions 1 and 2 and gains three neighbours that cost nothing, so $D_w(L_3) = D_w(L_2)$ exactly. We call L_3 the silent lift of L_2. The lift bounds the triplet optimum from above by the doublet optimum; the question is whether a triplet code can do strictly better.

Without chemistry it cannot. Setting $d = 1$ for every pair of distinct classes makes D_w count the non-synonymous neighbours per residue. Lindsey's (1964) edge-isoperimetric inequality for Hamming graphs gives, for any block A, at least $3|A|(n - \log_4|A|)$ edges leaving it. With equal usage and $\Sigma n_a = 4^n$, concavity of the logarithm then gives $D_w \geq 3 \log_4 k$ for every n. The bound depends on the number of classes and not on codon length, and it is attained by subcube blocks when k is a power of 4. Certified search reaches the bound at $k = 4$ and $k = 16$ (3.00 and 6.00) and finds the same optimum for doublets and triplets where the bound is not attained: 5.00 at $k = 8$ (bound 4.50) and 5.40 at $k = 10$ (bound 4.98) (table 1). Every one of 512 independent annealing runs reached the same value in each of these eight cells, with one exception (511 of 512 at $n = 3$, $k = 10$).

With chemistry the result holds under uniform usage and nearly holds under proteome usage. For the ten classes at uniform usage, the doublet and triplet optima are both 21.345. The best triplet code is the silent lift of the best doublet code: 58 of 64 searches started from the lift returned it, and 14 of 512 searches from random codes found the same value independently. At amino-acid frequencies from UniProtKB/Swiss-Prot (release 2026_03), the triplet optimum is 18.886 against 19.187 for the doublet, a difference of 1.6%. The triplet code gains this by giving the most frequent classes a few more codons than a lift allows (97% rather than 100% synonymy at the silenced position). A third position therefore does not, by itself, make a code more robust to misreading.

The invariance extends to the full alphabet. For 20 amino acids without chemistry the triplet and quadruplet optima are both 6.80 (bound $3 \log_4 20 = 6.48$), and with chemistry at uniform usage the best quadruplet code found is the silent lift of the best triplet code,

23.366 for both (table 3). These 20-amino-acid optima are less certain than the ten-class ones: two independent batches of 256 triplet searches disagreed by up to 0.7%.

Removing the three stop codons from the triplet graph lowers the best triplet code found to 19.95 (uniform usage) and 17.37 (proteome usage), at least 6.5% and 9.5% below the doublet optimum. This is not a robustness gain. A misreading into a stop codon has no cost in D_w, so a graph with stops simply counts fewer errors. A misreading into a stop codon truncates the protein, which is no less damaging than a missense substitution, so the comparison between architectures is made without stops.

A third position helps only as a discardable channel

The equality of the optima assumes every position is misread at the same rate. Misreading rates differ by position (Kramer and Farabaugh 2007; Katoh and Suga 2023; Landerer et al. 2024), and a triplet code can respond to that difference in a way a doublet code cannot: it can make its noisiest position silent and carry all information on the two most accurate ones.

For each of the seven published sets of position weights (w_1, w_2, w_3) and each of twelve graded settings (19 settings) under uniform usage, the triplet optimum equalled, to within 10^{-14}, the best doublet optimum over the two retained positions: $D_3^*(w_1, w_2, w_3) = \min_p D_2^*$(weights with position p removed), with the silenced position always the one with the largest weight. We have no proof of this equality, and under proteome usage it holds only approximately: the triplet code does better than the best lift by 0.006 to 0.30 units (at most 1.6%) through the same reallocation of codons as above.

The triplet advantage therefore depends on what a doublet decoder would do, which cannot be observed. We bracket it with two assumptions (fig. 4). If a doublet decoder read its two positions as accurately as positions 1 and 2 of a triplet decoder (the decoder-invariant case), a triplet code gains nothing when position 3 is the noisiest, whatever the ratio w_3/w_2, and gains only when one of positions 1 and 2 is noisier than position 3. If instead the last base read by a doublet decoder inherited the error rate of the wobble position (the terminal-sloppy case), the triplet advantage $1 - D_3^*/D_2^*$ rises with w_3/w_2: 31% at 2, 63% at 5, 79% at 10, 92% at 30 and 98% at 100 (uniform usage, $w_1 = w_2$).

Published misreading rates place the advantage between 0 and 32% (table 2). Two kinds of measurement disagree about which position is noisiest. In vitro, with the cognate amino acid withheld, near-cognate misincorporation by *Escherichia coli* ribosomes gives $w_1 : w_2 : w_3 = 0.98 : 1 : 1.39$ (Katoh and Suga 2023; our per-opportunity calculation from their Supplementary Fig. 8). Proteome-wide mass spectrometry in vivo gives 1.37 : 1 : 0.71 in *E. coli* and 1.40 : 1 : 0.71 in yeast (Landerer et al. 2024; our assignment of substitutions to mismatch positions from their Data S4). The optimal triplet code silences position 3 under the in vitro rates and position 1 under the in vivo rates. The in vivo rates cannot settle the matter, because they are measured in cells that use the SGC and they detect only amino-acid changes: most misreading at position 3 is synonymous in the SGC and therefore invisible, which depresses the apparent w_3. The in vitro rates measure what the decoding machinery confuses when the code does not hide the confusion.

The SGC silences position 3. Of single-nucleotide substitutions between sense codons, 71.6% are synonymous at position 3, 4.6% at position 1 and none at position 2. This is the structure the optimum predicts under decoder-intrinsic rates, and it identifies the robustness benefit of the triplet architecture as the cost of the discarded third channel: 0% if a doublet decoder would have read its second position well, 13 to 26% at the measured in vitro rates if its last base would have been read as loosely as the wobble position.

Why three and not four

Sixteen doublets cannot name twenty amino acids and a stop signal. That excludes $n = 2$ for the canonical alphabet, and the invariance above excludes any robustness reason for $n = 4$ at equal misreading rates. The discardable-channel result says where a fourth position could still help. A triplet code for 20 amino acids cannot make position 3 fully synonymous, because 16 first-two-position boxes are too few; the best codes leave 12 to 22% of position-3 substitutions non-synonymous. A quadruplet code has room to silence its loosest position completely and carry all information on three accurately read positions.

We computed the 20-amino-acid optima at $n = 3$ and $n = 4$ under two assumptions about the fourth position (table 3). If it were read as loosely as the wobble position, the quadruplet code gains nothing at the in vitro rates and at every graded ratio w_3/w_2: it silences one loose position and has no use for the second. If the ribosome monitored a third position as accurately as the first two, the best quadruplet codes we found gain the cost of the unsilenced part of the triplet's wobble channel: about 3 to 6% at the in vitro rates under uniform usage and 2 to 4% under proteome usage. These gains come from searches and could be somewhat larger at the true optimum (without physicochemical distances, the edge-isoperimetric bound limits any quadruplet gain to 4.7%). They rise to 20% only if position 3 were misread five times as often as position 2. Against that gain stands a certain cost: every residue would take four nucleotides of message and four positions of codon-anticodon pairing instead of three. Quadruplet decoding is not chemically out of reach: tRNAs of every isoacceptor class in *E. coli* can be converted to quadruplet decoders, although only nine of the twenty synthetases tolerate quadruplet anticodons (DeBenedictis et al. 2022). A structural analysis of fixed-length codebooks likewise identifies triplets as a Pareto choice (Lin et al. 2026, preprint). A code that is already within a few percent of the quadruplet optimum has no robustness reason to lengthen its codons, and the standard code's triplets sit at the shortest length that can name the alphabet.

The standard code sits 13 to 19% above the achievable minimum

The ten-class projection of the SGC, in which each amino acid is replaced by its class and the three stop codons are removed, costs $D_w = 25.38$ at uniform usage and 21.39 at proteome usage. The corresponding triplet optima are 21.34 and 18.89, so the SGC projection lies 19% and 13% above the best achievable code. The doublet projection of the SGC, obtained by majority vote over position 3, costs 22.07 and 19.57 on its nine classes (His loses the only prefix it contests, CA, in a tie). The SGC is thus far from the best reachable code under this objective, in line with the finding that improving moves remain available from it (Novozhilov et al. 2007).

Why codon-shuffling nulls overstate error minimization

A common alternative to the block-preserving null shuffles amino-acid labels among individual sense codons, keeping the number of codons per amino acid fixed (the codon shuffler of Rozhoňová and Payne 2021). It is a legitimate first choice, and it is the null we used in our initial analyses. Applied to doublet and triplet codes, it leads to the conclusion that triplet architecture enables deeper error minimization than doublet architecture. That conclusion is an artefact of the null. The null scatters synonymous codons across the Hamming graph, so any code whose synonyms are clustered scores far below it regardless of which amino acids are neighbours. The effect is large enough that chemistry becomes almost irrelevant: with every class equidistant from every other, the codon-shuffling null gives z = −17.07 for the SGC, −15.06 for the ten-class triplet projection and −3.66 for the doublet projection (10^6 shuffles each, seed 20260924), against −16.52, −14.16 and −3.62 with physicochemical distances on the same shuffles. Under this null the z-scores measure synonym clustering, and triplet codes have more synonyms to cluster, so comparing them across codon lengths reports a difference in synonymy rather than in error minimization.

The block-preserving null of Haig and Hurst (1991), which permutes amino acids among the SGC's synonymous blocks and keeps the blocks intact, isolates the assignment. Under it the SGC's noise distortion D has z = −2.34 (p = 0.010; 10^7 permutations) and its Dirichlet energy E, the sum of squared distances over all edges, has z = −1.90 (p = 0.027). The significance holds across property representations, including unscaled components, the 20 descriptors without principal component analysis and an earlier compiled descriptor matrix (p = 0.0005 to 0.027; table S3). For the doublet and ten-class triplet projections the block-preserving null is small enough to enumerate: 9! = 362,880 and 10! = 3,628,800 codes, configuration spaces within a factor of ten of each other. The doublet projection ranks in the top 0.5% of its space (p = 0.0051 on D), and the ten-class triplet projection is not significant (p = 0.17). Five classes have two members equally close to the class centroid, so the representatives are an arbitrary choice; across all 32 choices the doublet projection stays significant (p = 0.0001 to 0.017) and is always more significant than the ten-class triplet projection, whose p ranges from 0.009 to 0.20 (median 0.08; below 0.05 for 10 of 32 choices; table S1). The doublet projection's significance depends on rescaling the principal components (unscaled components or the 20 descriptors, p = 0.065 to 0.20; table S3), but the ordering of the two projections holds in every representation. Coarse-graining the SGC to ten classes removes much of its assignment signal, because the merged classes absorb the chemically closest neighbour pairs into synonymy.

Three consequences follow for comparisons of this kind. z-scores of observed codes against codon-shuffling nulls do not control for the size or the synonymy of the configuration space; a reduced-alphabet projection of the SGC is a coarse-graining of the SGC rather than an independent control; and within-architecture z-scores cannot separate architecture from assignment. This is why the architecture comparison in this paper rests on minima, which do not depend on any null.

Discussion

The genetic code does not become more robust to single-nucleotide misreading by being read in triplets. At fixed amino-acid usage the lowest achievable cost per residue is the same for doublets and triplets, exactly in the absence of chemistry and to within 1.6% with it, and the triplet and quadruplet optima for 20 amino acids coincide at uniform usage. The reason is geometric. A longer codon offers more codons per amino acid, which reduces the fraction of misreadings that change the amino acid, and more positions at which misreading occurs, which increases their number; the edge-isoperimetric bound shows that the two effects cancel for the optimal arrangement, and the silent lift shows how the cancellation is achieved.

A longer code gains only when positions are misread at different rates, because it can discard its worst channel. This quantifies the logic of Crick's wobble hypothesis (Crick 1966): a loosely read third position costs nothing if the code makes it synonymous, and the SGC does. Whether the triplet architecture gains anything over a doublet depends on a counterfactual that cannot be observed, the accuracy with which a doublet decoder would read its last base; the gain is zero if the decoding centre, which checks the geometry of the first two codon-anticodon base pairs (Ogle et al. 2001), would have checked both doublet positions as well, and 13 to 26% at the in vitro rates if it would not. The same arithmetic closes the question from the other side. A fourth position can gain only the unsilenced remainder of the wobble channel, and only if the ribosome monitored one more position; the best quadruplet codes we found gain 2 to 6% at the in vitro rates, and without chemistry the edge-isoperimetric bound limits any quadruplet gain over the best triplet code found to 4.7% (6.48 against 6.80); codons stop at the shortest length that names the alphabet because nothing in the error cost pays for a longer one.

Both results tie the benefit of the architecture to how each codon position is read. The table's optimum is the same at every codon length; what changes with length is the set of positions read loosely, which the code can therefore afford to silence, and which positions those are is measured from decoding rather than read off the table.

Three limits bound these conclusions. The cost counts single-nucleotide misreadings weighted by one distance matrix derived from 20 descriptors; the invariance without chemistry is independent of that choice, the qualitative conclusions hold for an earlier compiled descriptor matrix, for unscaled components and for the 20 descriptors without principal component analysis (table S3), but the sizes of the triplet and quadruplet advantages depend on the matrix. The optima with physicochemical distances are the best values found by annealing searches checked against the lift bound, certified in that sense rather than proven, and the 20-amino-acid optima in particular were reached by few runs. The analysis concerns misreading only; errors of aminoacylation are outside its scope.

The standard genetic code is a triplet code because triplets are the shortest words that name the alphabet, and it places its synonyms on the position its decoder reads loosely. Codon length itself buys no robustness: the gain of the triplet architecture is the cost of the channel it discards, and a fourth position would add at most a few percent.

Materials and methods

Amino-acid distances

Twenty-two side-chain properties were targeted, and each descriptor was taken from the AAindex1 database (Kawashima and Kanehisa 2000; file downloaded 2026-09-24) by accession, or computed from side-chain structure with RDKit (version 2026.03.1) by a stated rule; the rules for choosing entries were fixed before any statistic was computed. Sixteen are AAindex entries (table S2): hydropathy KYTJ820101, volume BIGC670101, polarity GRAR740102, isoelectric point ZIMJ680104, net charge KLEP840101, hydrogen-bond donors FAUJ880109, refractivity MCMT640101, bulkiness ZIMJ680102, helix, strand and turn propensities CHOP780201 to CHOP780203, mean fractional area loss ROSG850102, flexibility VINM940101, atom-based hydrophobic moment EISD860102, mean area buried on transfer ROSG850101, and molecular weight FASG760101. Four are computed from structure: hydrogen-bond acceptors, aromaticity, ring count and rotatable bonds. Two targeted properties had no source and were omitted: side-chain pKa (the four AAindex pK entries, FASG760104, FASG760105, FAUJ880113 and JOND750102, describe the α-amino and α-carboxyl groups) and a strand moment. No pair of descriptors correlated at $|r| \geq 0.95$, so all 20 were retained; two pairs lie within 0.01 of that threshold (fractional area loss and flexibility, $r = 0.943$; net charge and isoelectric point, $r = 0.941$), and a threshold of 0.94, which removes net charge and flexibility, leaves every conclusion unchanged (table S3). Each was standardized to zero mean and unit variance (population standard deviation). Principal component analysis by singular value decomposition retained eight components, the smallest number reaching 97% of the variance (97.5%). Each component's scores were then standardized across the 20 amino acids, which weights all eight components equally. Table S3 reports the main statistics with unscaled components, with the 20 standardized descriptors used directly, and with an earlier descriptor matrix compiled by hand from published scales, which gives the same qualitative conclusions. Distances are Euclidean in the rescaled eight-dimensional space.

Codes and classes

The SGC was taken from NCBI translation table 1. The ten classes are those of Murphy et al. (2000): {A, G}, {V, L, I, M}, {F, W, Y}, {S, T}, {N, Q}, {D, E}, {K, R}, {H}, {P}, {C}. Each class is represented by one member: A, L, W, S, N, D, R, H, P and C. For the five two-member classes both members are equidistant from their centroid, so the choice is arbitrary; table S1 reports the alternatives. The doublet projection assigns each first-two-position prefix the class held by the majority of its four triplets. Four prefixes are ties (AA, AG, CA, UU); they were broken by the order of the class identifiers X1 to X10 in table S1, which gives CA to the Asn/Gln class and leaves His without a prefix, so the projection has nine classes. No ordering of classes keeps all ten (ordering by amino-acid letter keeps His but loses Asn/Gln); breaking each tie toward the class that would otherwise lose its last prefix keeps all ten, and that projection gives $D = 19.97$ and a block-preserving p of 0.043 over its 10! codes (table S1).

Objective

D_w is defined in Results. Usage vectors were uniform ($f_a = 1/k$) or the sum of member amino-acid frequencies from UniProtKB/Swiss-Prot release 2026_03 (retrieved 2026-09-23), renormalized over the classes present. Two gates were checked before any search: D_w with $f_a = n_a/N$ reproduced, with the earlier descriptor matrix, D of the ten-class triplet projection (21.268639884706705) and of the doublet projection (18.937298834660997) computed by an independent implementation, to within 4×10^{-14}, and the silent lift preserved D_w to within 2×10^{-14} for the doublet projection and 100 random doublet codes under random usage vectors. The architecture comparison uses all 4^n codons as sense codons; the SGC projections exclude UAA, UAG and UGA.

Optimization and certification

Minima were found by simulated annealing over single-codon reassignments that keep every class non-empty, with geometric cooling and incremental evaluation, each run ending in steepest descent to a code that no single reassignment or pairwise swap improves. Each cell used 512 runs from random codes (256 in the position-weight analyses, 128 in the weight grid), and each triplet cell added runs started from all three single-position silent lifts of the best doublet codes. A triplet result was accepted only if it did not exceed the cost of every lift. The check did real work: in three weight-grid cells with $w_1 = 3$, 128 random runs ended above the optimum, and runs started from the lift that silences position 1 reached it; all 24 cells of the grid (twelve weight settings, two usage vectors) pass the check. Table 1 reports how many runs reached the best value in each cell.

Position-specific misreading rates

Rates per non-synonymous opportunity were computed from two primary sources. From Katoh and Suga (2023, Supplementary Fig. 8), each codon-amino-acid combination with a single mismatch was assigned to the mismatch position given in the figure, and the rate at each position is the fraction of tested combinations with detectable misincorporation. From Landerer et al. (2024, Data S4), each substitution from codon c to amino acid d was assigned to position p if a codon of d differs from c only at p; the rate at p is the number of such events divided by the codon exposure times the number of detectable non-synonymous single changes at p. Events requiring two or more changes, events ambiguous between positions, stop codons and I/L exchanges were excluded. Dataset-level bootstrap intervals and leave-one-dataset-out values are given in table S4.

Triplet and quadruplet codes

The 20-amino-acid comparison used each amino acid's own vector, all 4^n codons as sense codons, free block sizes and the same search, with 256 random runs at $n = 3$ and 128 random runs plus 40 lift-started runs at $n = 4$, each quadruplet lift built by inserting a silent position into the best triplet code for the remaining three weights. The fourth position was weighted either as a monitored position (weights w_1, w_2, w_2, w_3) or as a second loosely read position (w_1, w_2, w_3, w_3). Gates reproduced the SGC distortion computed with the earlier descriptor matrix (23.92236) to within 10^{-13}, and with the rebuilt matrix the chemistry-

blind control, which cannot depend on the matrix, reproduced z = −17.07 exactly and the lift invariance to 8×10^{-16}.

Null models

The codon-shuffling null permutes amino-acid labels among the 61 sense codons, preserving the number of codons per amino acid. The block-preserving null permutes amino acids among the SGC's synonymous blocks (Haig and Hurst 1991), with UCN and AGY treated as one serine block. For the SGC it was sampled 10^7 times; for the doublet and ten-class triplet projections it was enumerated exhaustively. With the rebuilt matrix the 10^7 permutations were drawn afresh (seed 20260924); the sampled null mean agreed with its closed-form value to within 0.93 (D) and 1.23 (E) standard errors. Reported p-values are the fraction of codes strictly better than the observed code.

Declarations

Funding

This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of interest

The authors declare no competing interests.

Author contributions

B.K. conceived the study and designed the analysis. H.E.R. built the Monte Carlo pipeline and figures. J.P. contributed to drafting and cross-disciplinary review. B.K. wrote the manuscript with contributions from H.E.R. and J.P. All authors approved the final version. [Kiran to confirm wording with co-authors.]

Data availability

All data analysed are published: amino-acid descriptors from AAindex (Kawashima and Kanehisa 2000), misincorporation rates from Katoh and Suga (2023, Supplementary Fig. 8), and substitution events from Landerer et al. (2024, Data S4). Derived tables are in the Supplementary Material and the archive below.

Code availability

All scripts, gates and outputs are archived at Zenodo, https://doi.org/10.5281/zenodo.21614179 (concept DOI, which resolves to the latest version) [add the new version DOI once published].

References

Błażej P, Kowalski D, Mackiewicz D, Wnetrzak M, Aloqalaa D, Mackiewicz P (2022) The structure of the genetic code as an optimal graph clustering problem. J Math Biol 85:9. https://doi.org/10.1007/s00285-022-01778-4

Crick FHC (1966) Codon-anticodon pairing: the wobble hypothesis. J Mol Biol 19:548-555. https://doi.org/10.1016/S0022-2836(66)80022-0

Crick FHC (1968) The origin of the genetic code. J Mol Biol 38:367-379. https://doi.org/10.1016/0022-2836(68)90392-6

DeBenedictis EA, Söll D, Esvelt KM (2022) Measuring the tolerance of the genetic code to altered codon size. eLife 11:e76941. https://doi.org/10.7554/eLife.76941

Drummond DA, Wilke CO (2008) Mistranslation-induced protein misfolding as a dominant constraint on coding-sequence evolution. Cell 134:341-352. https://doi.org/10.1016/j.cell.2008.05.042

Freeland SJ, Hurst LD (1998) The genetic code is one in a million. J Mol Evol 47:238-248. https://doi.org/10.1007/PL00006381

Haig D, Hurst LD (1991) A quantitative measure of error minimization in the genetic code. J Mol Evol 33:412-417. https://doi.org/10.1007/BF02103132

Katoh T, Suga H (2023) A comprehensive analysis of translational misdecoding pattern and its implication on genetic code evolution. Nucleic Acids Res 51:10642-10652. https://doi.org/10.1093/nar/gkad707

Kawashima S, Kanehisa M (2000) AAindex: amino acid index database. Nucleic Acids Res 28:374. https://doi.org/10.1093/nar/28.1.374

Koonin EV, Novozhilov AS (2009) Origin and evolution of the genetic code: the universal enigma. IUBMB Life 61:99-111. https://doi.org/10.1002/iub.146

Kramer EB, Farabaugh PJ (2007) The frequency of translational misreading errors in E. coli is largely determined by tRNA competition. RNA 13:87-96. https://doi.org/10.1261/rna.294907

Landerer C, Poehls J, Toth-Petroczy A (2024) Fitness effects of phenotypic mutations at proteome-scale reveal optimality of translation machinery. Mol Biol Evol 41:msae048. https://doi.org/10.1093/molbev/msae048

Lenstra R (2015) The graph, geometry and symmetries of the genetic code with Hamming metric. Symmetry 7:1211-1260. https://doi.org/10.3390/sym7031211

Lin R, Wang C, Hu Y, Wang C (2026) Why this code? A constrained mapping framework for the evolutionary stability of the canonical genetic code. bioRxiv. https://doi.org/10.64898/2026.08.28.747708

Lindsey JH II (1964) Assignment of numbers to vertices. Am Math Mon 71:508-516. https://doi.org/10.2307/2312587

Murphy LR, Wallqvist A, Levy RM (2000) Simplified amino acid alphabets for protein fold recognition and implications for folding. Protein Eng 13:149-152. https://doi.org/10.1093/protein/13.3.149

Novozhilov AS, Koonin EV (2009) Exceptional error minimization in putative primordial genetic codes. Biol Direct 4:44. https://doi.org/10.1186/1745-6150-4-44

Novozhilov AS, Wolf YI, Koonin EV (2007) Evolution of the genetic code: partial optimization of a random code for robustness to translation error in a rugged fitness landscape. Biol Direct 2:24. https://doi.org/10.1186/1745-6150-2-24

Ogle JM, Brodersen DE, Clemons WM Jr, Tarry MJ, Carter AP, Ramakrishnan V (2001) Recognition of cognate transfer RNA by the 30S ribosomal subunit. Science 292:897-902. https://doi.org/10.1126/science.1060612

Rozhoňová H, Payne JL (2021) Little evidence the standard genetic code is optimized for resource conservation. Mol Biol Evol 38:5127-5133. https://doi.org/10.1093/molbev/msab236

Seo Y, Tlusty T, Jo J (2026) The genetic code at the balance point of error and demand. PLoS Comput Biol 22:e1014613. https://doi.org/10.1371/journal.pcbi.1014613

Sonneborn TM (1965) Degeneracy of the genetic code: extent, nature, and genetic implications. In: Bryson V, Vogel HJ (eds) Evolving genes and proteins. Academic Press, New York, pp 377-397

Tlusty T (2007) A model for the emergence of the genetic code as a transition in a noisy information channel. J Theor Biol 249:331-342. https://doi.org/10.1016/j.jtbi.2007.07.029

UniProt Consortium. UniProtKB/Swiss-Prot release 2026_03 statistics. https://web.expasy.org/docs/relnotes/relstat.html (accessed 2026-09-23)

Woese CR (1965) On the evolution of the genetic code. Proc Natl Acad Sci USA 54:1546-1552. https://doi.org/10.1073/pnas.54.6.1546

Figure legends

Figure 1. Codes on the Hamming graph and the silent lift. (a) $H(2, 4)$ drawn as a 4 × 4 grid (rows, first base; columns, second base), and $H(3, 4)$ drawn as four copies of that grid, one per third base (after Lenstra 2015). Two codons are adjacent when they differ at one position: GCU (bold outline) has six neighbours in its own layer, at positions 1 and 2, and one neighbour in each of the other three layers, at position 3. (b) The doublet projection of the SGC (left; nine classes, coloured by class) and its silent lift (right): every layer of the triplet code is a copy of the doublet code, so position 3 is fully synonymous and every codon keeps the non-synonymous neighbours it had as a doublet. (c) The neighbourhood of GCU (Ala) in the SGC, with the physicochemical distance d to each neighbouring amino acid; position-3 neighbours are synonymous ($d = 0$). The per-residue cost D_w sums these distances over all neighbours, weighting each position by its misreading rate w_p and each amino acid by its usage f_a.

Figure 2. Codon-shuffling nulls measure synonym clustering. (a) z-scores of the noise distortion D of the doublet projection, the ten-class triplet projection and the SGC against the codon-shuffling null (10^6 shuffles, seed 20260924), with physicochemical distances

(blue) and with every pair of amino acids at the same distance (grey). (b) The same codes against the block-preserving null; the two projections were enumerated exhaustively (9! and 10! codes) and the SGC was sampled 10^7 times. p is the fraction of codes strictly better than the observed code. (c) Exhaustive block-preserving null distributions of D for the two projections, standardized to zero mean and unit variance; dashed lines mark the observed codes.

Figure 3. Codon length does not change the achievable cost. (a) Minimum number of non-synonymous neighbours per residue, without physicochemical distances, for doublets (circles) and triplets (squares) with k equally used classes, and for triplets and quadruplets with 20 classes (k = 20). The dashed line is the edge-isoperimetric bound $3 \log_4 k$, attained at k = 4 and 16. (b) Minimum per-residue cost D_w for ten classes with physicochemical distances under uniform (circles) and proteome (squares) usage, for the best doublet and triplet codes and for the two projections of the SGC.

Figure 4. The advantage of a longer code is the cost of the channel it discards. (a) Advantage of the best triplet code over the best doublet code, $1 - D_3^*/D_2^*$, against w_3/w_2 (with $w_1 = w_2$), under the terminal-sloppy and decoder-invariant assumptions; solid lines, uniform usage; dashed lines, proteome usage. Stars mark the in vitro misreading rates of Katoh and Suga (2023). (b) Advantage of the best quadruplet code over the best triplet code for 20 amino acids, when the fourth position is read as accurately as position 2 (monitored) or at the position-3 rate (loose); stars mark the in vitro rates. (c) Triplet advantage at the seven published rate sets of table 2 under the two assumptions about doublet decoding (uniform usage), with the position the best triplet code silences.

Tables

Table 1. Minimum per-residue misreading cost D_w for doublet (n = 2) and triplet (n = 3) codes at equal position weights. Hits: runs reaching the best value, as hits/runs; for triplets, random-start runs and then runs started from the silent lift of the best doublet code. The proteome-usage triplet search used 256 random starts. Without chemistry the lower bound $3 \log_4 k$ is 3.00, 4.50, 4.98 and 6.00 for k = 4, 8, 10 and 16; it is attained only at powers of 4.

Distances	Classes k	Usage	D_2^*	D_3^*	Triplet advantage	Hits n = 2	Hits n = 3 (random; lift)
none (d = 1)	4	uniform	3.000	3.000	0	512/512	512/512
none	8	uniform	5.000	5.000	0	512/512	512/512
none	10	uniform	5.400	5.400	0	512/512	511/512
none	16	uniform	6.000	6.000	0	512/512	512/512
physicochemical	10	uniform	21.345	21.345	0	458/512	14/512; 58/64
physicoc	10	proteom	19.187	18.886	1.6%	395/512	19/256;

Distances	Classes k	Usage	D_2^*	D_3^*	Triplet advantage	Hits n = 2	Hits n = 3 (random; lift)
hemical		e					9/64

Table 2. Triplet advantage $1 - D_3^*/D_2^*$ at published position-specific misreading rates, ten classes, physicochemical distances. Decoder-invariant: the doublet is read with the weights of triplet positions 1 and 2. Terminal-sloppy: the doublet's second position carries the position-3 weight.

Rate source	$w_1 : w_2 : w_3$	Position silenced	Advantage, decoder-invariant (uniform / proteome)	Advantage, terminal-sloppy (uniform / proteome)
Katoh and Suga 2023, in vitro, strict	0.98 : 1 : 1.39	3	0 / 0.9%	15.2% / 13.4%
Katoh and Suga 2023, intensity-weighted	0.99 : 1 : 1.79	3	0 / 0	26.0% / 22.6%
Landerer et al. 2024, *E. coli*	1.37 : 1 : 0.71	1	28.0% / 28.8%	15.9% / 15.0%
Landerer et al. 2024, *E. coli*, largest dataset excluded	1.91 : 1 : 1.38	1	16.3% / 15.3%	27.6% / 28.3%
Landerer et al. 2024, yeast	1.40 : 1 : 0.71	1	28.8% / 29.4%	17.0% / 15.9%
Landerer et al. 2024, yeast, largest dataset excluded	1.16 : 1 : 0.57	1	29.1% / 32.2%	8.2% / 8.7%
equal rates	1 : 1 : 1	any	0 / 1.6%	0 / 1.6%

Table 3. Twenty amino acids: best triplet (D_3^*) per-residue cost and quadruplet advantage $G = 1 - D_4^*/D_3^*$, uniform / proteome usage. Monitored: the fourth position read as accurately as position 2. Loose: read at the position-3 rate. Two independent batches of triplet searches disagreed by up to 0.7% (uniform) and 1.2% (proteome); D_3^* is the lower, and G is uncertain by about ±1 percentage point.

Triplet weights	D_3^*	G, monitored	G, loose
none (d = 1), equal	6.800 / not	0 / not computed	0 / not computed

Triplet weights	D_3^* computed	G, monitored	G, loose
equal (1 : 1 : 1)	23.366 / 21.990	0 / 0	0 / 0
Katoh strict (0.98 : 1 : 1.39)	23.854 / 22.300	3% / 2%	0 / 0
Katoh intensity (0.99 : 1 : 1.79)	24.630 / 22.731	6% / 4%	0 / 0
graded (1 : 1 : 2)	24.939 / 23.067	6% / 5%	0 / 0
graded (1 : 1 : 5)	29.189 / 26.143	20% / 16%	0 / 0
graded (1 : 1 : 10)	35.243 / 31.330	34% / 30%	0 / 0

Supplementary material

Supplementary tables. Table S1, representative-member sensitivity. Table S2, the 20 descriptors. Table S3, main statistics under six descriptor representations (the rebuilt and the earlier compiled matrix, each as rescaled or unscaled principal components or as the 20 standardized descriptors) and under a correlation threshold of 0.94. Table S4, position-specific rate calculation, bootstrap intervals and leave-one-dataset-out values. Table S5, the weight grid of the terminal-sloppy and decoder-invariant analyses.